

Transformers, Attention, and All That

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ECE 657A

Outline

- **Failures of RNNs**
- The Attention Mechanism
 - Using Attention to Understanding CNNs
 - Learning Attention Masks to Improve CNN Performance
- Rise of the Transformer Age
 - Attention replaces Recurrence
 - BERT all the BERTs

Limitations of RNNs and LSTMs for NLP

- See other slidedeck “Attention Is All You Need Presentation.pdf” on the original transformer attention paper
- But first, what is attention anyways?

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What is Attention Anyways?

- *Background:* So, we hear about attention or the attention mechanism being an important part of many new ML advances, particularly ***Transformers***.
- **But what is attention?**
- When I started making these slides last year, I thought I knew what “attention” was...
 - so I looked around and found out that everyone is very sure they understand it,
 - and also very sure that the latest version they are using is the *only* one.
- But the *idea* of looking at the attention (or focus, or “internal thinking”)
 - of a DNN to *understand* and *improve* its learning has been around for a while

Self-Supervised Representation Learning

- Some of the ideas of how to do self-supervised learning for Natural Language were introduced in 2011
 - “Natural Language Processing (Almost) from Scratch”
[Collobert and Weston \(JMLR, 2011\)](#)
- Using end-to-end DNN to learn encodings for each word
- Minimum text pre-processing
- Many of these ideas made way into Word2Vec, BERT and others

The Original Use of Attention in CNNs

Using an attention calculation on an already trained CNN to indicate to the user what features the CNN is using to make its classification. It also ties these attention scores to descriptive word labels.

Bengio, ICML, 2015 - Show, Attend and Tell
<http://proceedings.mlr.press/v37/xuc15.pdf>

Using Attention to make CNNs Interpretable

- It was very popular for a few years
- Grad-CAM and other algorithms created cool images shows us what CNNs are “thinking”.
- But they were computed after a CNN was learned, so it didn’t really say anything about how it relates to the learning process.

Using Attention to make CNNs Interpretable

- Also, some people have pointed out that even these methods can highlight things that aren't really there:
- *Use HiResCAM instead of Grad-CAM for faithful explanations of convolutional neural networks* - Rachel Lea Draelos, Lawrence Carin - <https://arxiv.org/abs/2011.08891>
- *CNN Heat Maps: Class Activation Mapping (CAM)* - Rachel Draelos - <https://glassboxmedicine.com/2019/06/11/cnn-heat-maps-class-activation-mapping-cam/> (June 11, 2019)
 - **Discuss on Hypothesis:** <https://hyp.is/go?url=https%3A%2F%2Farxiv.org%2Fpdf%2F2004.09666.pdf&group=xrxZM1b3>

Learning Attention Masks to Improve CNN Performance

- **Blog Explainer:** This is a good overview article for many issues about attention as it was used originally in explainability of CNNs.
 - “Learn to Pay Attention! Trainable Visual Attention in CNNs” - Rachel Draelos
<https://towardsdatascience.com/learn-to-pay-attention-trainable-visual-attention-in-cnns-87e2869f89f1> (Aug 10, 2019)
- **Paper:** Torr, ICLR, 2018 - “Learn to Pay Attention”
<https://arxiv.org/pdf/1804.02391.pdf>

Learning with Attention for Images Classification and Labelling

Another application of this approach:

**Data Efficient and Weakly Supervised
Computational Pathology on Whole Slide Images** -
Lu, Nature Biomedical Engineering, 2021

Arxiv paper : <https://arxiv.org/pdf/2004.09666.pdf>

Discuss on hypothesis :

<https://hyp.is/go?url=https%3A%2F%2Farxiv.org%2Fpdf%2F2004.09666.pdf&group=xrxZM1b3>

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New “Age” of Attention with Transformers

Seminal Paper

- Vaswani, et. al. “*Attention is All you Need*”. Conference on Neural Information Processing Systems (NeurIPS). Long Beach, CA, 2017.
- **Arxiv Paper** : <https://arxiv.org/pdf/1706.03762.pdf>
- **Discuss on hypothesis** : <https://hyp.is/go?url=https%3A%2F%2Farxiv.org%2Fpdf%2F1706.03762.pdf&group=xrxZM1b3>

What about Recurrent Links?

“In this work we propose the Transformer, a model architecture **eschewing recurrence** and instead **relying entirely on an attention mechanism** to draw **global dependencies** between input and output.”

[Attention is All you Need -
Vaswani, NeurIPS 2017]

- So *instead* of having links to previous “states” of the network - a transformer uses time as a special kind of feature where we are looking for dependencies of varying scales - and it using the **attention mechanism** to achieve that

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BERT all the BERTs

Two steps

1. pre-training

- model is trained on *unlabeled data*
- over different pre-training *tasks*

2. fine-tuning -

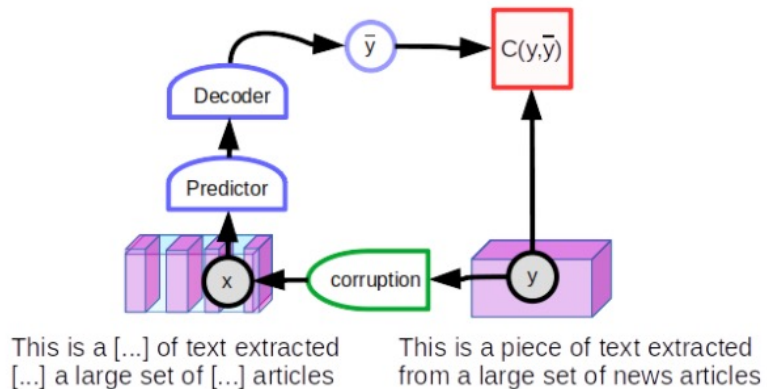
- first, the model is *initialized* with the pre-trained parameters,
- Then all model the parameters are *fine-tuned* using *labeled data* from the downstream tasks

BERT Definitions

- Input: sequence of words
- Masked auto-encoder:
 - a type of denoising auto-encoder
 - mask out some percentage of words (~15%)
- Predict:
 - missing words only (or distribution over likely words)
 - a Denoising Auto-Encoder would try to predict all of the output completely
 - this saves a lot of learning
- Original Paper:
 - Hypothesis annotated link:
<https://hyp.is/go?url=https%3A%2F%2Farxiv.org%2Fpdf%2F1810.04805.pdf&group=world>

BERT Model

A masked language model, which is an instance of denoising auto-encoder, itself an instance of contrastive self-supervised learning. Variable y is a text segment; x is a version of the text in which some words have been masked. The network is trained to reconstruct the uncorrupted text.



<https://ai.facebook.com/blog/self-supervised-learning-the-dark-matter-of-intelligence/>

Ongoing History of AIML

Transformers are the new CNN (or SVM (or GMM (or Bayes Net (or Prolog Model))))))

- Nice history of recent developments leading to Transformers at
 - https://huggingface.co/docs/transformers/model_summary
 - Hypothesis links:
https://hyp.is/go?url=https%3A%2F%2Fhuggingface.co%2Fdocs%2Ftransformers%2Fmodel_summary&group=world
- BERT – encoder only, bidirectional
- GPT – decoder only, single directional

- Recent CBC Spark podcast on ethics and impact of ChatGPT on Society, Philosophy, Culture
- <https://www.cbc.ca/radio/spark/bots-like-chatgpt-aren-t-sentient-why-do-we-insist-on-making-them-seem-like-they-are-1.6761709>
- Jill Fellows is a philosophy instructor at Douglas College in British Columbia

What impact will Chat-GPT etc have on society?

Jill Fellows is a philosophy instructor at Douglas College in British Columbia

- Chat-GPT seems like like it's our servant to us
 - but it is also learning from us...or, its owners are
 - Continued pattern of services using AI/ML in exchange for data, advertising, money
- Ethics of AI
- the meaning of consciousness
- The interaction between these tools and our culture
- Future of Life Institute:
 - Pause the Experiment!
 - <https://futureoflife.org/open-letter/pause-giant-ai-experiments/>

Why Chatbots?

Chatbots have been one of the main focusses of AI research even before the word "chatbot" exists.

- Back to the original Turing Test, remember?
- In the 70's, AI researchers created a therapist chatbot called Eliza